

Cinema Outings AI

Multi-Agent Autonomous Architecture, A2UI Protocol & Cloud Infrastructure

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Submission Track: Google AI L200	Region: us-central1 (Council Bluffs)	Infrastructure as Code: Terraform >= 1.5.0

Executive Summary: Cinema Outings AI is a production-grade, full-stack multi-agent mobile and cloud platform designed to discover cinema screenings, personalize recommendations using taste memory, guarantee transaction safety, book tickets, and synchronize calendar invites. Built using the **Google Agent Development Kit (ADK)**, **Gemini 2.5** strategic tiered model routing, **FastMCP** (Model Context Protocol), and Flutter's dynamic **A2UI (Agent-to-UI)** declarative protocol, the architecture is provisioned with enterprise-grade **Terraform** Infrastructure as Code (IaC) on Google Cloud Platform.

1. Architectural Foundations & Design Principles

The architecture is engineered around five fundamental enterprise principles:

- 1. Supervisor-Host Orchestration:** Built upon Google ADK's LlmAgent pattern. The Outing Coordinator Agent acts as the central host, delegating sub-tasks to specialized sub-agents while maintaining unified conversational state and session memory passing.
- 2. Declarative Agent-to-UI (A2UI) Protocol:** Rather than returning raw markdown or brittle plain text, agents output typed A2UI JSON components. The Flutter client dynamically parses and renders rich interactive widgets (movie discovery carousels, 2D seat selectors, boarding pass ticket passes, and calendar invitations).
- 3. Defense-in-Depth & Sensitive Data Protection:** Strict Google ADK lifecycle plugins inspect user input before agent execution. Google Cloud Sensitive Data Protection (DLP) automatically redacts payment cards, phone numbers, emails, and SSNs. Reservation safety plugins ensure zero unheld ticket charges.
- 4. Strategic Tiered Model Routing:** Semantic intent classification routes each request to its cost-optimal Gemini model tier—reserving gemini-2.5-pro for nuanced taste reasoning, gemini-2.5-flash for conversational agility and deterministic booking transactions, and gemini-2.0-flash-lite for profile maintenance and calendar formatting.
- 5. Fully Codified Infrastructure as Code (IaC):** All Google Cloud resources—including Cloud Run v2, Secret Manager, Artifact Registry, Cloud Storage FUSE volumes, Cloud Monitoring, and least-privilege IAM service accounts—are provisioned automatically via modular Terraform configurations.

Architecture Blueprint Navigation: Overleaf (Page 2) features the complete graphical system architecture diagram depicting the six synchronized layers down to agent, tool, protocol, and cloud service components. Pages 3 and 4 detail agent specifications, tool signatures, security guardrails, model tiers, and Terraform IaC.

Figure 1: Full-System Architecture & Component Interaction Blueprint

High-resolution component diagram detailing the interaction flow across Client, Gateway, Security, Multi-Agent Core, Tools & Cloud Persistence:

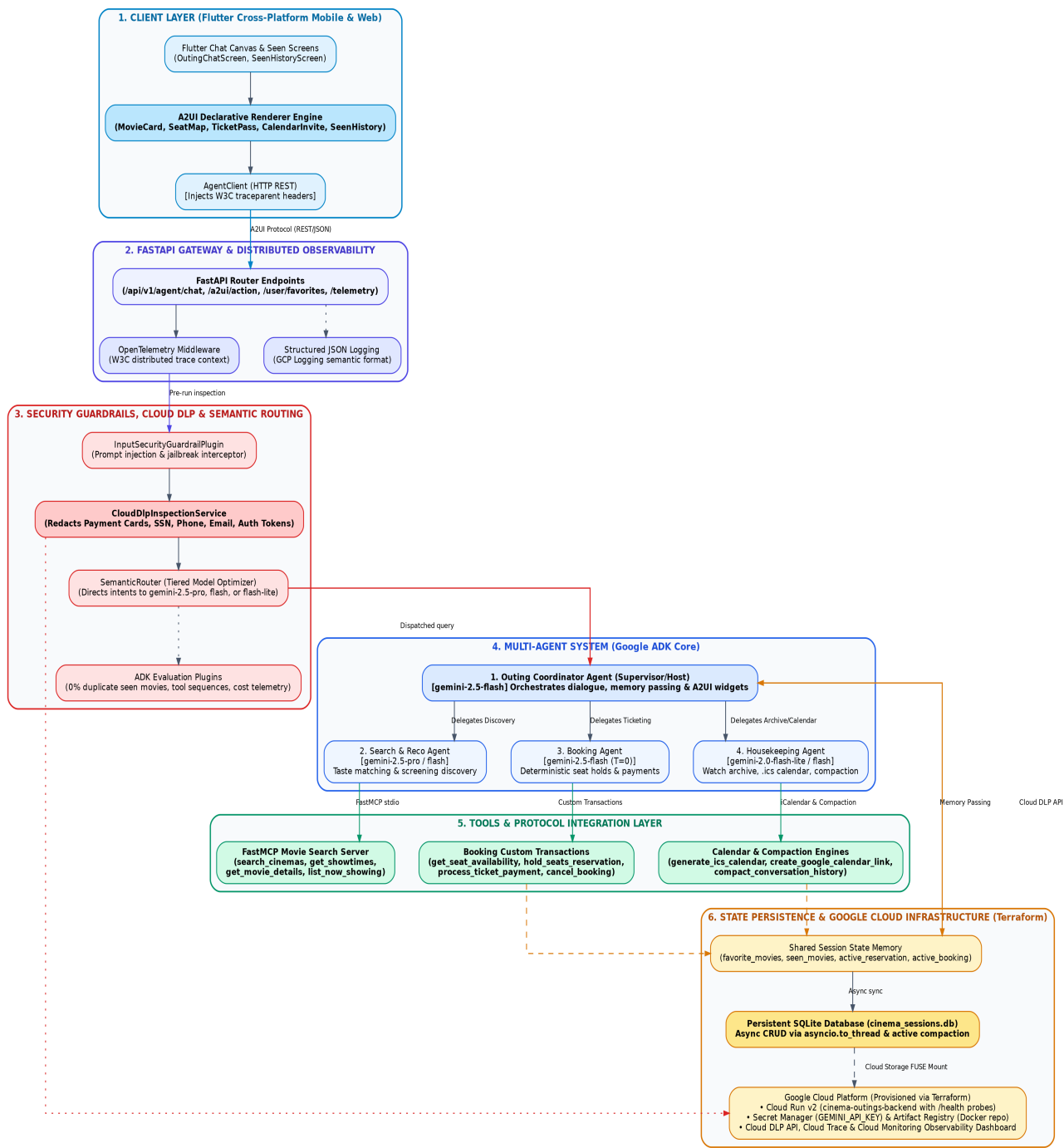


Figure 1 Legend: Blue: Client & ADK Agent Core | Purple: FastAPI Gateway & Telemetry | Red: Security Guardrails & Cloud DLP | Green: FastMCP & Transaction Tools | Amber: Session State, SQLite & GCP Cloud Run v2 (Terraform).

2. Specialized Multi-Agent System Breakdown

Each agent executes a specialized role within clear isolation boundaries, coordinated via shared Google ADK session memory:

Agent Role	Technology & Model	Core Architectural Responsibilities	Session Memory Keys
1. Outing Coordinator (Supervisor/Host)	Google ADK gemini-2.5-flash	• Conversational intent classification • Sub-agent routing & delegation • A2UI declarative widget packaging • Dialog lifecycle & turn history	conversation_history active_intent session_id
2. Search & Reco	Google ADK + FastMCP gemini-2.5-pro (Deep) gemini-2.5-flash (Fast)	• Live screenings lookup via FastMCP • Director & genre taste-matching • 0% duplicate seen movie filtering • Venue, hall & showtime discovery	Reads: favorite_movies seen_movies
3. Booking Agent	Google ADK + Custom Tx gemini-2.5-flash (T=0, Deterministic)	• Real-time seating grid inspection • 10-minute temporary seat reservation • Payment settlement & idempotency • Digital QR token boarding pass	Mutates: active_reservation active_booking
4. Housekeeping	Google ADK + Summarizer gemini-2.0-flash-lite gemini-2.5-flash	• Watched archive ledger maintenance • User favorites list curation • RFC-5545 .ics & Google Calendar export • Automatic conversation compaction	Mutates: seen_movies favorite_movies summary

3. Tools, Protocols & Integration Specifications

Standardized tool contracts connect the multi-agent core with cinema catalogs, transactional APIs, and client widgets:

Protocol / Subsystem	Implementation File	Methods & Interface Signatures	Operational Guarantees
FastMCP Server (Model Context Protocol)	backend/mcp_servers/ movie_search_server.py	• search_cinemas(location, query) • get_showtimes(movie_title, date) • get_movie_details(movie_title) • list_now_showing(genre)	FastMCP stdio server providing screening catalogs, showtimes, auditorium formats (IMAX, standard), and capacity limits.
Ticketing Engine (Custom Transactions)	backend/tools/ booking_transactions.py	• get_seat_availability(c, h, t) • hold_seats_reservation(...) • process_ticket_payment(...) • cancel_booking(booking_id)	Atomic reservation management. Implements lock-lease timeouts (600s), idempotency keys, subtotal calculations, and digital QR boarding passes.
Calendar Engine	backend/tools/ calendar_tools.py	• generate_ics_calendar(...) • create_google_calendar_link(...)	RFC-5545 iCalendar specification generator (`.ics`) and browser-compatible Google Calendar deep links with cinema venue and seat data.
A2UI Protocol (Agent-to-UI)	backend/protocols/ a2ui.py	• A2UIMessage • movie_card, seat_map_selector • ticket_pass, calendar_invite_card • seen_history_list	Declarative JSON protocol enabling autonomous backend agents to construct native Flutter widgets dynamically without app recompilation.
Compaction Engine	backend/state/ history_summarizer.py	• compact_conversation_history(...) • extract_key_entities(...)	Autonomous conversation compaction. Compresses aged dialogue turns into running entity summaries while keeping recent turns intact.

4. Security Guardrails, Cloud DLP & Observability

The system enforces defense-in-depth across the agent execution lifecycle using Google ADK BasePlugin hooks, OpenTelemetry distributed tracing, and Google Cloud Sensitive Data Protection (DLP):

Subsystem	Implementation File	Security & Operational Guarantees
Cloud DLP (Sensitive Data)	backend/security/dlp_service.py	Inspects & redacts sensitive PII before agent processing: Credit Cards ([REDACTED_PAYMENT_CARD]), Emails, Phone Numbers, Social Security Numbers, Bearer/API Tokens.
Input Security Guardrail	backend/plugins/security_guardrails.py	Intercepts prompt injection attempts ('Ignore previous instructions', role hijacking), jailbreak sequences, and malicious formatting attacks before model execution.
Booking Safety Guardrail	backend/plugins/security_guardrails.py	Enforces that payments cannot execute without an active, unexpired seat hold token. Blocks replay attacks, double-charges, and seat reservation concurrency collisions.
OpenTelemetry Tracing	backend/telemetry/tracing.py	W3C traceparent HTTP header propagation across Flutter client, FastAPI gateway, LLM inference, MCP tool calls, and SQLite persistence.
Structured JSON Logging	backend/telemetry/logging.py	GCP Cloud Logging semantic format with severity levels, trace correlation (logging.googleapis.com/trace), span IDs, and session context metadata.

5. Strategic Model Routing & Terraform Infrastructure (IaC)

Semantic intent classification routes queries to optimal Gemini model tiers, all provisioned via modular Terraform:

Model Tier	Model Identifier	Assigned Agents & Roles	Architectural & Cost Rationale
Deep Reasoning	gemini-2.5-pro	Search & Reco (Taste Analysis)	Nuanced stylistic comparisons, director filmography synthesis, and complex multi-criteria recommendation grounding.
Fast Conversational	gemini-2.5-flash	Coordinator, Search (Catalog)	Sub-second intent classification, conversational turn handling, and high-throughput screening discovery.
Deterministic Tx	gemini-2.5-flash (T=0)	Booking Agent	Strict schema compliance, zero-temperature execution for reservation holding and financial transaction calls.
Ultra-Lite	gemini-2.0-flash-lite	Housekeeping & Calendar	Lowest cost tier for conversation compaction, profile memory updates, and simple RFC-5545 `.ics` formatting.

Modular Terraform Infrastructure as Code (GCP):

- **modules/apis:** Enables 14 GCP APIs (Cloud Run, Cloud DLP, Vertex AI, Secret Manager, Cloud Trace, Logging, Monitoring).
- **modules/iam:** Creates dedicated least-privilege service account (cinema-outings-sa) with strict role bindings.
- **modules/secrets:** Manages GEMINI_API_KEY in Google Secret Manager with automated accessor policies.
- **modules/storage:** Provisions encrypted GCS bucket with versioning & lifecycle rules, mounted to Cloud Run via FUSE.
- **modules/registry:** Private Artifact Registry Docker repository for container images.
- **modules/cloud_run:** Cloud Run v2 service deploying the FastAPI multi-agent backend with startup/liveness health probes.
- **modules/monitoring:** Cloud Monitoring dashboard (latencies, CPU/RAM, requests, Cloud DLP PII log metrics) and 5xx alerts.
- **Environments:** Pre-configured environments/dev (cost-optimized) and environments/prod (high-availability warm instances).

Automated Verification & Test Results: All 63 pytest test cases pass cleanly across agent delegation, memory passing, A2UI protocol serialization, MCP tool execution, security guardrails, Cloud DLP sanitization, OpenTelemetry tracing, conversation compaction, and Terraform syntax/module validation.